DNA finger printing involves identifying differences in some specific regions in DNA sequence called **repetitive DNA**, because in these sequences, a small stretch of DNA is repeated many times. These repetitive DNA are separated from bulk genomic DNA as different peaks during density gradient centrifugation. The bulk DNA forms a major peak and the other small peaks are referred to as **satellite DNA**. Depending on base composition (A : T rich or G : C rich), length of segment and number of repetitive units, the satellite DNA is classified into many sub categories such as micro-satellites, mini-satellites, etc., These sequences do not code for any proteins, but they form a large portion of human genome. These sequences show high degree of polymorphism and form the basis of DNA fingerprinting **(Fig. 5.15)**. DNA isolated from blood, hair, skin cells, or other genetic evidences left at the scene of a crime can be compared through VNTR patterns, with the DNA of a criminal suspect to determine guilt or innocence. VNTR patterns are also useful in establishing the identity of a homicide victim, either from DNA found as evidence or from the body itself.

The Steps in DNA Fingerprinting technique is depicted in Fig. 5.16.

1. **Extraction of DNA**

   The process of DNA fingerprinting starts with obtaining a sample of DNA from blood, semen, vaginal fluids, hair roots, teeth, bones, etc.,

2. **Polymerase chain reaction (PCR)**

   In many situations, there is only a small amount of DNA available for DNA fingerprinting. If needed many copies of the DNA can be produced by PCR (DNA amplification).

3. **Fragmenting DNA**

   DNA is treated with restriction enzymes which cut the DNA into smaller fragments at specific sites.

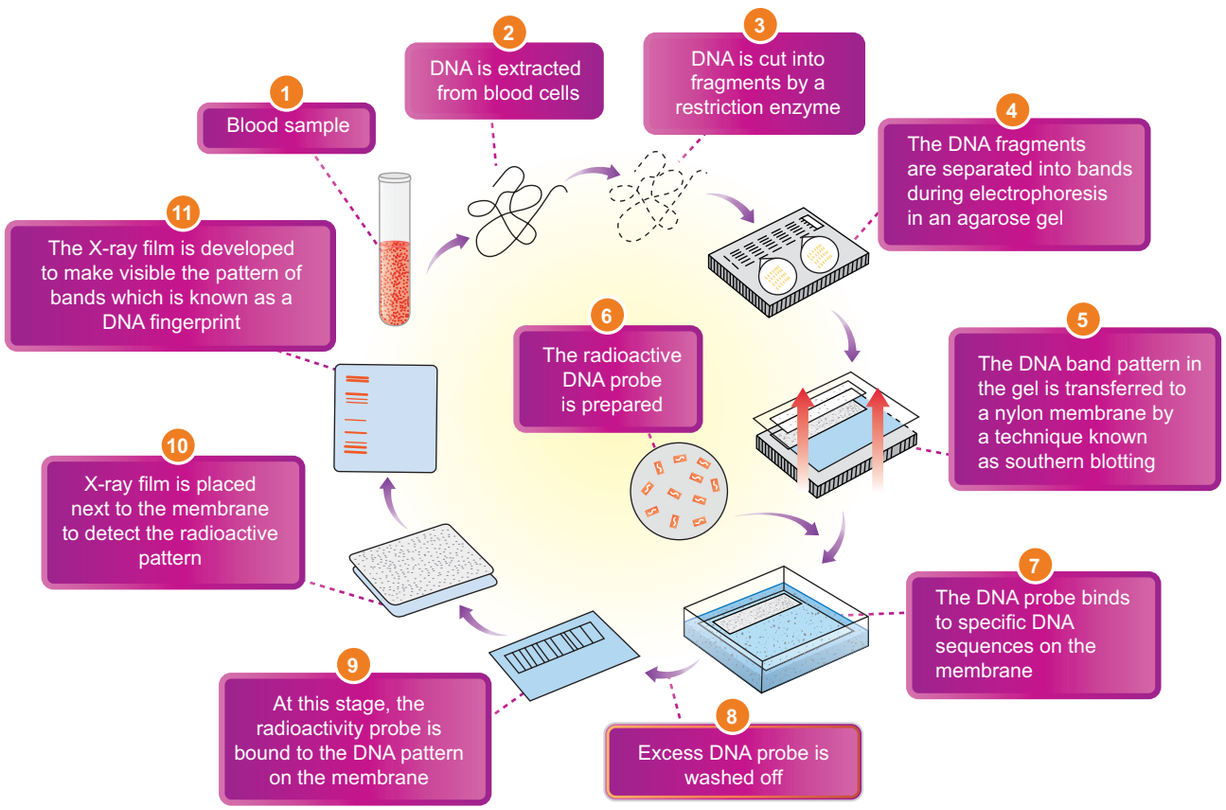


Fig. 5.16 Steps in DNA finger printing

TN_GOVT_XII_Zoology_chapter5.indd 85 24-02-2020 4.01.39 PM